

EXP.NO-7	Write a Servlet program to store a user's preferences using cookies. Retrieve and display these preferences on subsequent visits

AIM:

To create a Java Servlet program that uses cookies for session tracking in the TechScope product comparison website.

ALGORITHM:

- 1. Create an HTML form to collect user preferences
- 2. Submit form data to PreferenceServlet
- 3. In servlet, create cookies for theme and language
- 4. Set cookie expiry time
- 5. Add cookies to response
- 6. Redirect to display servlet
- 7. Retrieve cookies using request.getCookies()
- 8. Display stored preferences

CODE:

preferences.html

```
<!DOCTYPE html>

<html>

<head>

    <title>TechScope Preferences</title>

    <link rel="stylesheet" href="style.css">

</head>

<body>

<div class="card">

    <h1>Set Preferences</h1>

    <form action="PreferenceServlet" method="post">

        <label>Theme:</label><br>

        <select name="theme">
```

```
<option value="dark">Dark</option>

<option value="light">Light</option>

</select><br><br>

<label>Language:</label><br>

<select name="language">

    <option value="English">English</option>

    <option value="Tamil">Tamil</option>

</select><br><br>

<button type="submit">Save Preferences</button>

</form>

</div>

</body>

</html>
```

PreferenceServlet.java

```
package com.techscope;

import jakarta.servlet.*;
import jakarta.servlet.http.*;
import java.io.*;

import jakarta.servlet.annotation.WebServlet;

@WebServlet("/PreferenceServlet")

public class PreferenceServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    protected void doPost(HttpServletRequest request, HttpServletResponse response)

        throws ServletException, IOException {

        String theme = request.getParameter("theme");

        String language = request.getParameter("language");
```

```

        Cookie themeCookie = new Cookie("theme", theme);

        Cookie langCookie = new Cookie("language", language);

        themeCookie.setMaxAge(60 * 60 * 24); // 1 day

        langCookie.setMaxAge(60 * 60 * 24);

        response.addCookie(themeCookie);

        response.addCookie(langCookie);

        response.sendRedirect("PreferenceDisplayServlet");

    }
}

```

PreferenceDisplayServlet.java

```

package com.techscope;

import jakarta.servlet.*;
import jakarta.servlet.http.*;
import java.io.*;

import jakarta.servlet.annotation.WebServlet;

@WebServlet("/PreferenceDisplayServlet")

public class PreferenceDisplayServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        String theme = "dark";

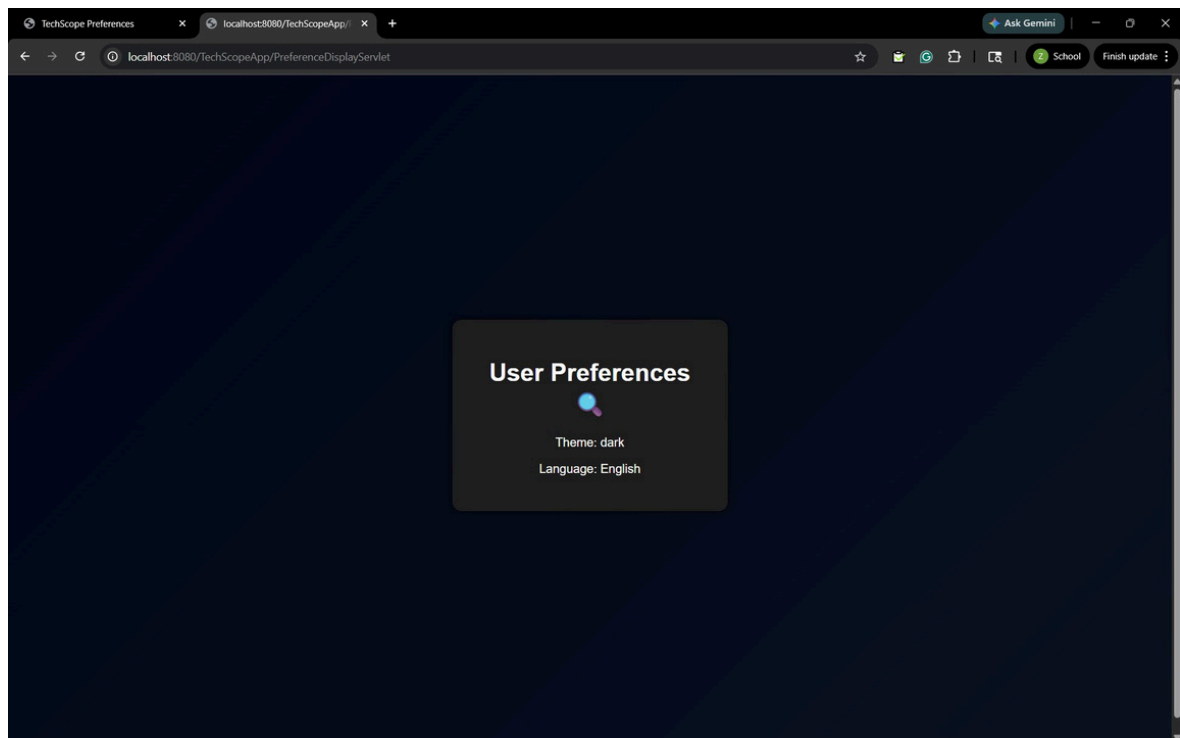
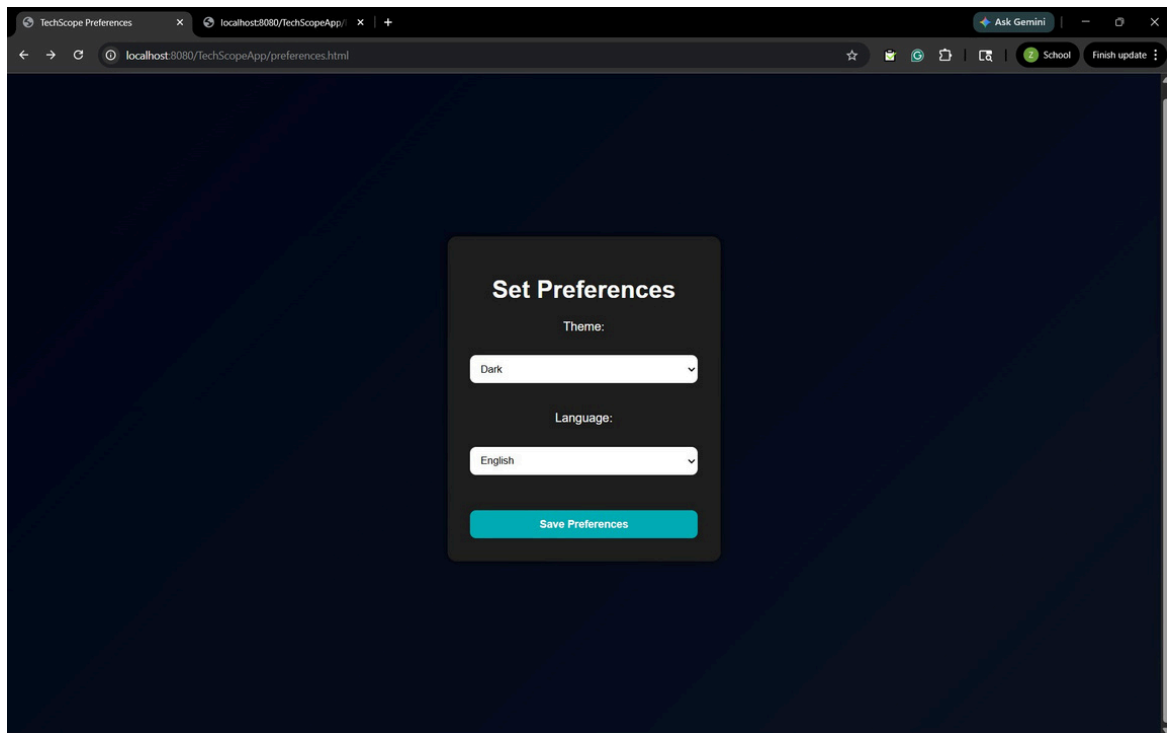
        String language = "English";

        Cookie[] cookies = request.getCookies();

```

```
if (cookies != null) {  
    for (Cookie c : cookies) {  
        if (c.getName().equals("theme")) {  
            theme = c.getValue();  
        }  
        if (c.getName().equals("language")) {  
            language = c.getValue();  
        }  
    }  
}  
  
out.println("<html><head><link rel='stylesheet' href='style.css'></head><body>");  
out.println("<div class='card'>");  
  
out.println("<h1>User Preferences    </h1>");  
out.println("<p>Theme: " + theme + "</p>");  
out.println("<p>Language: " + language + "</p>");  
  
out.println("</div>");  
out.println("</body></html>");  
}  
}
```

OUTPUT:



RESULT:

Thus, User preferences are stored in cookies and retrieved successfully on subsequent visits.